

ASSIGNMENT - 3

(1)

$$(A) \quad L_{core,1} = \frac{N^2}{R_{g,core,1}} = \frac{N^2 \mu_0 A_{g,core,1}}{2 l_{g,core,1}}$$

Assuming $A_{g, \text{core}} = A_{c, \text{core}} = 40 \text{ mm}^2$

$$\therefore L_{\text{core 1}} = \frac{100 \times 100 \times 4\pi \times 10^{-7} \times 40 \times 10^{-6}}{2 \times 1 \times 10^{-3}}$$

$$L_{core} = 251.325 \mu H$$

Saturation Current $I_{sat, core} :-$

$$B_{sat} = \frac{\phi}{A_c} \quad \text{and} \quad \phi = \frac{NI}{R_g} = \frac{NI}{\frac{2lg}{\mu_0 \mu_r}} \quad \text{--- (2)}$$

using ① and ②

$$B_{\text{sat}} = \frac{NI \mu_0 A_g}{2 l_g} \cdot \frac{1}{A_c} = \frac{NI_{\text{sat, core 1}} \mu_0}{2 l_g}$$

$$\therefore I_{\text{sat, core 1}} = \frac{2 \lg B_{\text{sat}}}{N \mu_0} = \frac{2 \times 1 \times 10^{-3} \times 0.2}{100 \times 4\pi \times 10^{-7}}$$

$$I_{\text{sat, core}} = 3.18 \text{ A}$$

(B) Similarly

$$L_{\text{core 2}} = \frac{N^2 \mu_0 A_{g, \text{core 2}}}{2 l_{g, \text{core 2}}}$$

$$= \frac{100 \times 100 \times 4\pi \times 10^{-7} \times 100 \times 10^{-6}}{2 \times 5 \times 10^{-3}}$$

$$L_{\text{core 2}} = 125.66 \mu\text{H}$$

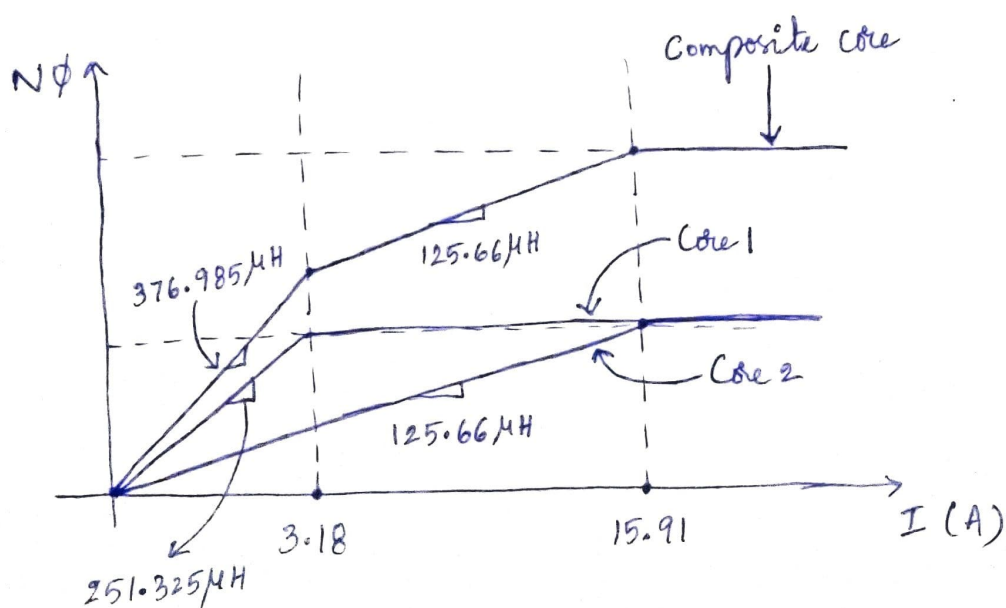
similarly

$$I_{\text{sat, core 2}} = \frac{2 l_{g, \text{core 2}} B_{\text{sat}}}{N \mu_0}$$

$$= \frac{2 \times 5 \times 10^{-3} \times 0.2}{100 \times 4\pi \times 10^{-7}}$$

$$I_{\text{sat, core 2}} = 15.91 \text{ A}$$

(C)



(2) (A) Core area $A_c = 20 \times 20 = 400 \text{ mm}^2$

(B) Window area $A_w = (15 + 15) \times 10 = 300 \text{ mm}^2$

(C) Maximum Energy stored:

$$\text{From } A_c A_w \geq \frac{L i_{pk} i_{rms}}{B_m J K_w}$$

$$L i_{pk} i_{rms} \leq A_c A_w B_m J K_w$$

$$\frac{1}{2} L i_{pk} i_{rms} \leq \frac{1}{2} A_c A_w B_m J K_w$$

$$\therefore E_{max} \leq \frac{1}{2} A_c A_w B_m J K_w$$

$$E_{max} = \frac{1}{2} \times 400 \times 10^{-6} \times 300 \times 10^{-6} \times 1.2 \times 4 \times 10^6 \times 0.5$$

$$E_{max} = 144 \text{ mJ}$$

(D) From above $E_{max} = 144 \times 10^{-3} \text{ J}$

$$\frac{1}{2} L_{max} i_{pk} i_{rms} = 144 \times 10^{-3}$$

Assuming no ripple $i_{pk} = i_{rms} = 1 \text{ A}$

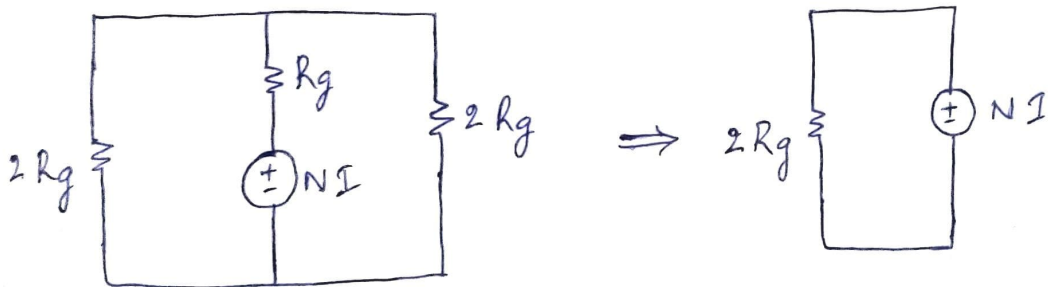
$$\therefore L_{\max} = 288 \text{ mH}$$

$$(E) \quad L_{\max} = \frac{N\phi}{I_{pk}} \Rightarrow N = \frac{L_{\max} I_{pk}}{B_{\max} A_c}$$

$$\therefore N = \frac{288 \times 10^{-3} \times 1}{1.2 \times 400 \times 10^{-6}}$$

$$N = 600 \text{ turns}$$

To find l_g :-



$$\therefore L = \frac{N^2}{2R_g} = \frac{N^2 \mu_0 A_g}{2l_g}$$

$$\therefore l_g = \frac{N^2 \mu_0 A_g}{2L_{\max}} = \frac{(600)^2 \times 4\pi \times 10^{-7} \times 400 \times 10^{-6}}{2 \times 288 \times 10^{-3}}$$

$$\therefore l_g = 0.314 \text{ mm}$$

(F) To check fringing $l_g \leq \sqrt{A_c}$

$$\therefore 0.314 \text{ mm} \leq 20 \text{ mm}$$

$\therefore$ Fringing effect is absent.

(3) Given $f = 25 \text{ kHz}$, $J = 2 \times 10^6 \text{ A/m}^2$

$$B_{pk} = 0.1 \text{ T}, K_w = 0.5$$

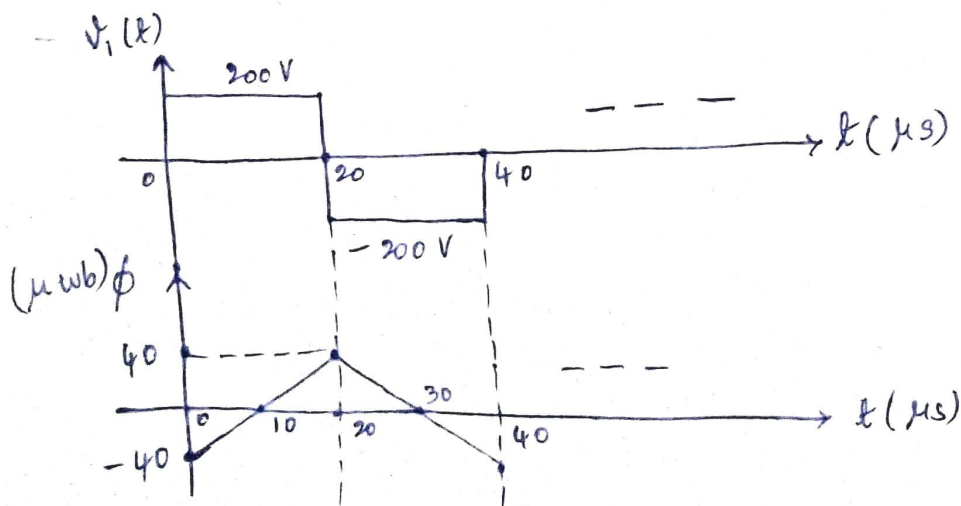
$$A_c = 20 \times 20 = 400 \text{ mm}^2 \text{ and } A_w = (15+15) \times 10 = 300 \text{ mm}^2$$

$$(A) VA_{max} = \frac{1}{2} 4 f B_{pk} A_c J K_w A_w$$

$$= \frac{1}{2} \times 4 \times 25,000 \times 0.1 \times 400 \times 10^{-6} \times 2 \times 10^6 \times 0.5 \times 300 \times 10^{-6}$$

$$\therefore VA_{max} = 600 \text{ VA}$$

(B)



$$(C) \quad V_{1, \text{rms}} = 4 f N_1 B_{pk} A_c$$

$$\therefore N_1 = \frac{V_{1, \text{rms}}}{4 f B_{pk} A_c} = \frac{200 \times 10^6}{4 \times 25,000 \times 0.1 \times 400}$$

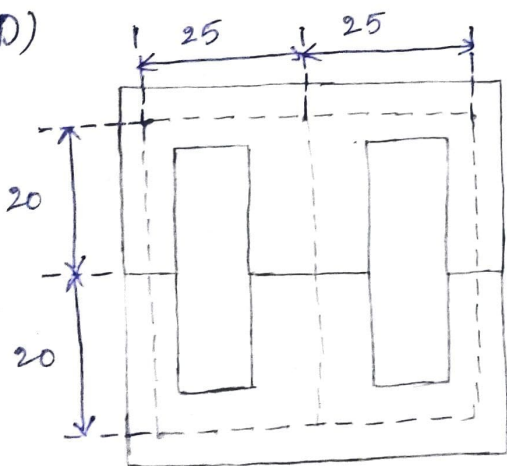
$$N_1 = 50 \text{ turns}$$

Similarly,

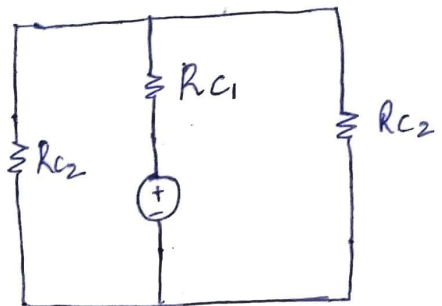
$$N_2 = \frac{V_{2, \text{rms}}}{4 f B_{pk} A_c} = \frac{16}{4 \times 25,000 \times 0.1 \times 400 \times 10^{-6}}$$

$$N_2 = 4 \text{ turns}$$

(D)



$\Rightarrow$

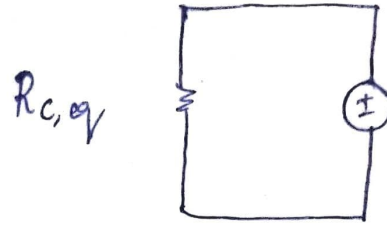


$$R_{C1} = \frac{l_1}{\mu_0 \mu_r A_{C1}} = \frac{40 \times 10^{-3}}{4\pi \times 10^{-7} \times 4000 \times 400 \times 10^{-6}}$$

$$= 19894.36 \text{ AT/Wb}$$

$$R_{C2} = \frac{l_2}{\mu_0 \mu_r A_{C2}} = \frac{90 \times 10^{-3}}{4\pi \times 10^{-7} \times 4000 \times 200 \times 10^{-6}}$$

$$R_{c2} = 89524.655 \text{ AT/Wb.}$$

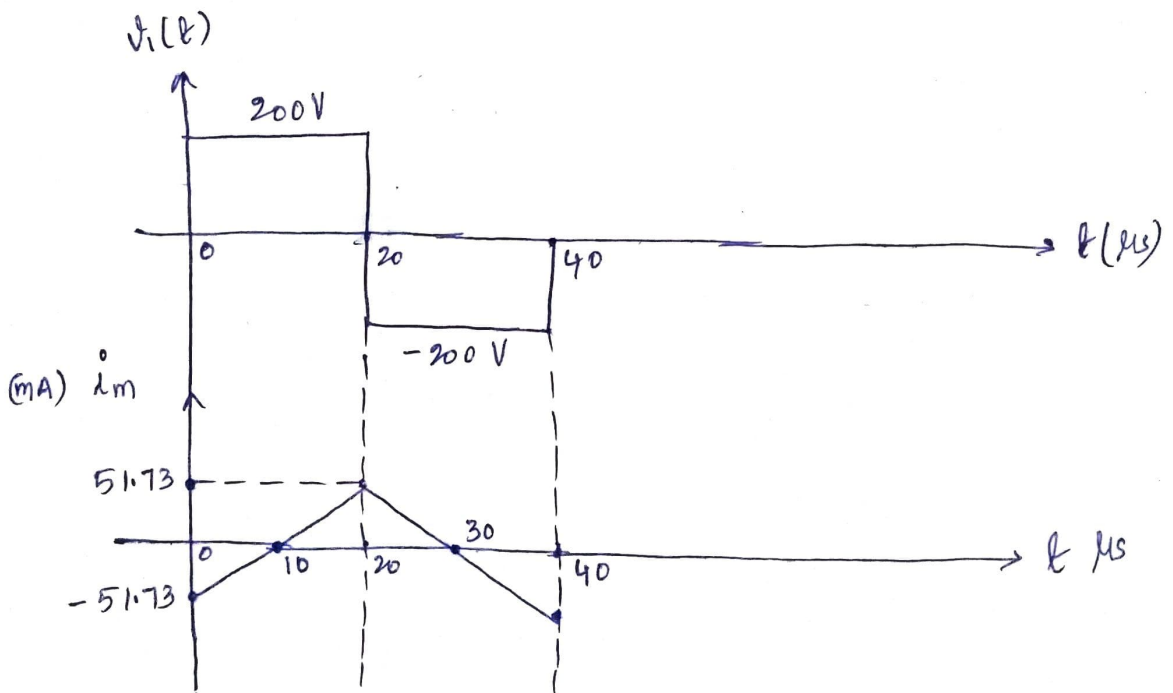


$$\therefore R_{c,eq} = R_{c1} + \frac{R_{c2}}{2} = 64,656.687 \text{ AT/Wb}$$

$$\therefore L_m = \frac{N_1^2}{R_{c,eq}} = \frac{50 \times 50}{64,656.687}$$

$$L_m = 38.66 \text{ mH}$$

(E)



$$L_m = \frac{N \phi}{I_m} \Rightarrow I_{m,pk} = \frac{N_1 B_{pk} A_c}{L_m}$$

$$= \frac{50 \times 0.1 \times 400 \times 10^{-6}}{38.66 \times 10^{-3}}$$

$$\therefore I_{m,pk} = 51.73 \text{ mA}$$